

Proposed Solution to #5816 SSMJ

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Statement of the Problem: For $0 < a \leq b$, prove:

$$\int_a^b \frac{xe^{x^2}\sqrt{e^{x^2}}}{e^{3x^2}+1} dx \leq \frac{1}{2} \tan^{-1} \left(\frac{e^{b^2}-e^{a^2}}{1+e^{a^2+b^2}} \right).$$

Solution of the Problem: Observe that :

$$\begin{aligned} \frac{1}{2} \tan^{-1} \left(\frac{e^{b^2}-e^{a^2}}{1+e^{a^2+b^2}} \right) &= \frac{1}{2} \left(\tan^{-1}(e^{b^2}) - \tan^{-1}(e^{a^2}) \right) = \frac{1}{2} (\tan^{-1}(e^{x^2}))|_a^b \\ &= \frac{1}{2} \int_a^b \frac{d}{dx} (\tan^{-1}(e^{x^2})) dx = \int_a^b \frac{xe^{x^2}}{1+e^{2x^2}} dx \end{aligned}$$

Now, the problem reduces to proving

$$\int_a^b \frac{xe^{x^2}\sqrt{e^{x^2}}}{e^{3x^2}+1} dx \leq \int_a^b \frac{xe^{x^2}}{1+e^{2x^2}} dx$$

for which it suffices to prove

$$\frac{\sqrt{e^{x^2}}}{e^{3x^2}+1} \leq \frac{1}{1+e^{2x^2}}$$

Since the denominators are clearly positive, we can cross multiply

$$e^{\frac{x^2}{2}} + e^{\frac{5x^2}{2}} \leq e^{3x^2} + 1 \quad (1)$$

To prove this statement, we use Karamata Inequality.

Since sequence $(3, 0) \succ (\frac{5}{2}, \frac{1}{2})$ and $f(a) = e^{ax^2}$ is convex whenever a is positive, thus by Karamata inequality,

$$f(3) + f(0) \geq f\left(\frac{5}{2}\right) + f\left(\frac{1}{2}\right)$$

Using $f(a) = e^{ax^2}$ proves equation (1) and hence the original inequality.